THE HALTING PROBLEM OF THE FOUR TURING MACHINES CM16/26/51/52: CRITERIA, ERA ALGEBRA, CORRESPONDENCE FACTS, AND THE OPEN PROBLEM

1. The four machines and conventions

Machine	Transition table	Halting transition	Sole in-edge(s) of the halting state
cm16	`1RB0LE_0RC1RB_1LC1RD_1RF0LA_0RD1LE_1RB---`	F on 1	$D0 =$ `1RF`
cm26	`1RB1RC_1LC0LE_1RD1RC_1LB0RA_---0LF_1RD1LF`	E on 0	$B1 =$ `0LE`
cm51	`1RB1RA_1LC0RC_1RE0LD_0RE0LF_1RA---_1RB1LF`	E on 1	$C0 =$ `1RE`, $D0 =$ `0RE`
cm52	`1RB0LD_1LC1RD_0LE1LA_0RB1RD_---0LF_1RA1LF`	E on 0	$C0 =$ `0LE`

Bi-infinite tape, initially all 0, head at cell 0, state A. The fourth column was checked row-by-row against the transition tables (the transitions into the halting state are exactly those listed, no others). The four halting transitions occupy different positions; criteria are established independently per machine (§2), followed by the era algebra and the era-level correspondence (§3), the verified correspondence facts (§4), and the per-machine reduction and open problem (§5–§6). The criterion layer is exhaustively verified by `casecheck-halt.mjs` (204,800 configurations, 111,257 halting instances, zero counterexamples, entry enumeration ENUM-OK ×4); the routine-layer island assertions are verified by `casecheck-island.mjs` (39 assertions, all passing, **cm16 only** — see the domain statement of Theorem 6.2).

2. Halt criteria (independent per machine)

Theorem 2.1 (cm16)**.** *cm16 halts if and only if at some time state D reads a 0 at cell w with $\tau(w{+}1) = 1$.*

Theorem 2.2 (cm26)**.** *cm26 halts if and only if at some time state B reads a 1 at cell q with $\tau(q{-}1) = 0$.*

Theorem 2.3 (cm51)**.** *cm51 halts if and only if at some time state C or D reads a 0 at cell w with $\tau(w{+}1) = 1$.*

Theorem 2.4 (cm52)**.** *cm52 halts if and only if at some time state C reads a 0 at cell c with $\tau(c{-}1) = 0$.*

cm16. The halting transition is F reading 1 ($F1$ undefined). The sole in-edge of F is $D0 =$ `1RF`: D reads 0 at cell w, writes 1 there, moves right to $w{+}1$ entering F; so every arrival of F immediately follows a $D0$, and the cell F reads is exactly $w{+}1$. Hence the machine halts iff after some $D0$ we have $\tau(w{+}1) = 1$, which is the stated criterion. (The trigger condition $\tau(w) = 0$ of $D0$ is contained in "D reads a 0 at w".) Both directions follow from this single in-edge chain. □

cm26. The halting transition is E reading 0. The sole in-edge of E is $B1 =$ `0LE`: B reads 1 at cell q, writes 0 there, moves **left** to $q{-}1$ entering E; the cell E reads is $q{-}1$. Halting $\iff$ after some $B1$ we have $\tau(q{-}1) = 0$. □

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cm51. The halting transition is E reading 1. The in-edges of E are exactly two: $C0 =$ `1RE` and $D0 =$ `0RE`, both moving **right** (after writing, to $w{+}1$ entering E). Hence halting $\iff$ after some $C0$ or $D0$ we have $\tau(w{+}1) = 1$. □

cm52. The halting transition is E reading 0. The sole in-edge of E is $C0 =$ `0LE`: C reads 0 at cell c, moves left to $c{-}1$ entering E, and E reads $\tau(c{-}1)$. □

Remark 2.5. *The four proofs are purely transition-level (enumerating one or two in-edges) and use no dynamics; the role of the checker is to cross-check the statement of the criteria against all instantiated configurations (guarding against typos), not a necessary part of the proofs.*

3. Lap grammar and era algebra

Definition 3.1 (marker states and eras)**.** *A marker state is the (A, h, τ) at the end of a dance-family routine; the evolution between consecutive markers is one era. The blank-tape initial segment enters the first marker by direct simulation.*

Lemma 3.2 (machine-specific lap words, one per machine)**.** *From* each machine's own canonical anchor *(the realization ρ_M of Definition 5.1 in §5), the (state, symbol) event word and period of one lap (round) — on the drain domain Rge4 (the cm16 row also covering the casecheck grid Rge2) — with R = lap-start reservoir and the final event the pivot boundary event shared with the next round, are:*

Machine	*round event word*	*period*
cm16	$A0\,B0\,C1\,D1\,A1\,E0\,D0\,F0\,B1^{R-2}\,B0\,C0\,C0\,C1\,D1\,A1\,E1^{R-1}\,E0\,D1\;(+A0)$	$2R{+}13$
cm26	$D0\,B1\,E1\,F1^{R-2}\,F0\,D1\,A1\,C1^{R-4}\,C0\;(+D0)$	$2R{+}1$
cm51	$A0\,B0\,C1\,D1\,F1^{R-1}\,F0\,B1\,C1\,D0\,E0\,A1^{R-3}\;(+A0)$	$2R{+}5$
cm52	$C0\,E1\,F1^{R-1}\,F0\,A1\,D1\,D0\,B1\,D1^{R-3}\,D0\,B0\;(+C0)$	$2R{+}5$

The net effect is $(x, R, z) \mapsto (x{+}1,\ R{-}2,\ z{+}1)$ for all four machines. The cm16 row is the word in the paper's standard configuration (left seam 0^2, A on the seam's first cell, right reservoir 1^R, right seam 0^2, deposit 1^z): $2R{+}14$ events in total (the final $A0$ being the boundary event, the round proper $2R{+}13$ transitions), the working window shifting one cell right with the head in the same relative position.

Proof. Complete derivation for the cm16 row; rows 2–4 are machine-verified (below).

Let the left seam be cells $g, g{+}1$ (values $0, 0$), the right reservoir 1^R occupying $g{+}2..g{+}R{+}1$, the right seam $g{+}R{+}2, g{+}R{+}3$, the deposit 1^z from $g{+}R{+}4$. Head A at g.

Segment 1 (8 steps): $A0$=`1RB` writes $1@g$, right; $B0$=`0RC` at $g{+}1$; $C1$=`1RD` at $g{+}2$; $D1$=`0LA` writes $0@g{+}3$, left; $A1$=`0LE` writes $0@g{+}2$, left; $E0$=`0RD` at $g{+}1$; $D0$=`1RF` writes $1@g{+}2$, right; $F0$=`1RB` writes $1@g{+}3$, right. Now $g{+}2, g{+}3 = 1, 1$, head B at $g{+}4$ (inside the reservoir). Between steps 7 and 8 the tape shows a transient island ($g{+}1, g{+}2, g{+}3 = 0, 1, 0$) with head distance ≤ 1.

Segment 2 ($R{+}4$ steps): $B1$=`1RB` keeps writing 1 moving right across $g{+}4..g{+}R{+}1$ ($R{-}2$ cells); $B0$ at $g{+}R{+}2$ (`0RC`); $C0\,C0$ = two applications of `1LC` writing 1 at $g{+}R{+}3$ then $g{+}R{+}2$ (moving left); $C1$ at the reservoir end $g{+}R{+}1$ (`1RD`); $D1$ writes $0@g{+}R{+}2$; $A1$ writes $0@g{+}R{+}1$, moving left into E at $g{+}R$.

Segment 3 ($R{+}2$ steps): $E1$=`1LE` keeps writing 1 moving left back across $g{+}R..g{+}2$ ($R{-}1$ cells); $E0$=`0RD` at $g{+}1$; $D1$=`0LA` writes $0@g{+}2$, ending at A on $g{+}1$ — the start of the next round.

Net effect check: the final tape has $g{=}1$ (left block +1); new left seam $g{+}1, g{+}2$; reservoir remnant $g{+}3..g{+}R$ ($R{-}2$ cells); new right seam $g{+}R{+}1, g{+}R{+}2$; $g{+}R{+}3{=}1$ absorbed into the deposit (+1). That is $(x{+}1,\ R{-}2,\ z{+}1)$, with the next round's head on the left cell of the new left seam — same relative position, window shifted by one. Step count $2R{+}14$. The word and net effect were re-verified verbatim on the parameter grid $R \in \{13, 20, 40\}$ by `casecheck-island.mjs` (cm16). Rows

2–4: the machine-specific words from each canonical anchor were verified verbatim on the parameter grid $R = 4..21$ and several (L, Z) environments (`ref-words-param.mjs`/`ref-words-small.mjs`, equivalence-referee session); the periods are pinned independently by the EV intervals of `quad6.exe` (R=8/13/20/21 → cm16 29/39/53/55, cm26 17/27/41/43, cm51/cm52 21/31/45/47). □

Remark 3.3 (from the paper's standard configuration the other three machines do *not* execute the cm16 word)**.** *The table requires each machine's* own *canonical-anchor geometry. If one starts instead from the paper's standard configuration (the cm16 geometry: left seam* 0^2*, A on the left seam), the third and fourth events are forced by the transition table to deviate: on cm26* $C1$ = `1RC` *goes to C (not D), on cm52* $C1$ = `1LA` *goes to A, and on cm51* $D1$ = `0LF` *goes to F (not A) — the three machines deviate at the 3rd/4th event already (independent check* `ref-lapword.mjs`*: the actual words begin* $A0\,B0\,C1\,C1\,C1\cdots$ *for cm26,* $A0\,B0\,C1\,D1\,F1\,F1\cdots$ *for cm51, and* $A0\,B0\,C1\,A1\,D1\,D0\,B1\cdots$ *for cm52). The correct common layer of the four machines is the* era level *(the drain-map level): Theorem 5.2 in §5.*

Remark 3.4 (era structure (verified facts))**.** *The per-lap bookkeeping of cm16 is given by Lemma 3.2 (right reservoir* -2 *per lap). Reservoir exhaustion triggers the drain and the churn reset; the reset-branch key sets (key = (position class, neighborhood block count)) collected from deep scans: cm16/26/51 share 78 keys identically, cm52 has 89. The drain layer is an affine cascade (Theorem 5.2); the churn/reseed layer has no closed form (table-level evidence of Proposition 5.3).*

Definition 3.5 (eras and resets)**.** *One era = the lap chain up to reservoir exhaustion + the drain routine + the reset. The working triple* (x, R, z) *evolves along the chain by Lemma 3.2 (cm16; the other machines at their own pivot events, Lemma 3.2 and Theorem 5.2); draining triggers a churn cascade (reseeding). Era key sets (key = (position class, neighborhood block count)) collected from deep scans: cm16/26/51 share 78 keys identically; cm52 has 89.*

4. The era system: data and structure

Definition 4.1 (reservoir configuration)**.** *The marker tape consists of: a left reservoir* 1*-block, a right reservoir* 1*-block and a deposit block, separated by* 0^2 *seams, with the frozen skeleton as tail (identical bit-for-bit across the four machines, fixed by deep scans); the working triple* (x, R, z) *records the working-block, right-reservoir and deposit lengths.*

Remark 4.2 (era-chain example (cm16, near 10^9 steps))**.** *The measured working-triple sequence of eras:*

$$(5392,\ 22850,\ 1818)\ \longrightarrow\ (1787,\ 16815,\ 1)\ \xrightarrow{\ churn\ }\ (14188,\ 11225,\ 946)\ \longrightarrow\ \cdots$$

The right reservoir decreases by steps of 2 along the lap chain until exhaustion, triggering the drain and the churn cascade (reseeding into a new reservoir layout); the chain truncations and reset branches form the era key set.

Remark 4.3 (cross-machine alignment data)**.** *The large-era* (L_0, R_0) *sequences of the four machines align over 562 measured pairs (remeasured as* $(\Delta L, \Delta R)$ *vector offsets: m26↔m16* $\in \{(1,0){:}556, (0,2){:}6\}$*, m51↔m16* $\in \{(1,-1){:}556, (0,1){:}6\}$*, m52↔m16 constant* $(0,2)$*, each with 6 flip points) with zero misalignment (prefix of the* 3×10^9*-step era census); the era key sets are 78 keys, identical key-by-key, for cm16/26/51 and 89 for cm52 (the shared 78 plus 11 right-edge keys). These are observations, not used as theorems; moreover the table-level comparison proves this alignment to be a* trajectory phenomenon*, not a consequence of drain-layer identity (Proposition 5.3(iii)).*

5. Correspondence between the four machines: the drain-layer theorem and three boundaries

Definition 5.1 (abstract marked states and canonical realization)**.** *An abstract marked state is* $s = (b, \mathrm{pos})$*: b lists all* 1*-block lengths and* pos *indexes the working triple (the abstraction of an*

era-anchor event*; churn-phase pivot events lie outside this domain). The canonical realization* $\rho_M(s)$ *separates blocks by* 0^1 *except inside the triple, where the seam widths are machine constants* $cm16{=}(2,2)$, $cm26{=}cm52{=}(1,1)$, $cm51{=}(1,2)$*; the head and state are the machine's anchor event (*$cm16$*: A0 at the L|R seam start;* $cm26{=}D0$, $cm51{=}A0$, $cm52{=}C0$ *at the R|Z seam start). Fidelity evidence: 8 real anchors (S1's five plus the equivalence referee's three new ones, at depths* 2,000,090/777,816/123,526*) with rebuild == real (era records field-by-field) plus event-class scans (400 pivot-event windows per machine: zero violations on era-anchor events) [*`quad-equiv.mjs` *S1 +* `ref-fidelity.mjs`*; see dossier §15 / attack-quad.md §15].*

Theorem 5.2 (drain-layer correspondence)**.** *Let* $F_M(s)$ *denote the map from* $\rho_M(s)$ *to the exhaustion of the current era, with output the lap count* ν *and the closing triple* (eL, eR, eZ)*. On the shared domain* $R \geq 4$*:*

1. $F_{cm51} = F_{cm52} = F_{cm16}$ *field-by-field (and also at* $R = 3$*);*
2. $F_{cm26} = \sigma \circ F_{cm16}$*, where the correction* σ *acts only on odd R:* $\nu_{26} = \nu_{16} - 1$ *and* $(eL, eR, eZ)_{26} = (eL - 1,\ eR + 2,\ eZ - 1)_{16}$*;*
3. *the terminal residues are* $eR \in \{1, 2\}$ *(*$cm16/51/52$*) or* $\{2, 3\}$ *(*$cm26$*), and always* $eL = L_0 + \frac{R_0 - eR}{2}$, $eZ = Z_0 + \frac{R_0 - eR}{2}$*.*

$R = 3$ *is a structural divergence point (*3 *is already* $cm26$*'s odd terminal residue, so* $cm26$ *has no drain continuation, while the other three run one further lap to* 1*); for* $R \leq 2$ *the map enters the churn layer, outside this theorem. The correspondence is abstract: the lap periods differ per machine (*$2R{+}13/2R{+}1/2R{+}5/2R{+}5$*, R = lap-start reservoir; Lemma 3.2).*

evidence grade (machine-checked). (i) whole-orbit: over the 3×10^9-step era census the identity of (3) holds exactly for all 166,976 *complete* eras of the four machines (one cap-truncated snapshot per machine excluded, four in all); (ii) synthetic table: of the 390 rows (78 shared keys × 5 parameter sets), the 110 all-clean rows contain 90 non-exhausted-input rows identical (with σ) 90/90 (the remaining 20 rows have $R \leq 2$ exhausted inputs, outside the shared domain); (iii) counterfactual replay of real-orbit states: 30/30 on $R \geq 4$. Independent referee re-run (`ref-sigma.mjs`): even R 220/220 identical, odd R 220/220 via σ (R up to 439), $R = 3$ divergent 20/20. □

Proposition 5.3 (three boundaries of the correspondence)**.** (1) *(churn layer) the eraB (post-reseed) component is identical on* 0/110 *clean shared rows and* 0/40 *real states;* $cm51$*'s key changed in all* 150 *observed cases (largest value deviation* $(-991, -598, 0)$*; unboundedness unproved), while* $cm26$ *agrees pairwise on* 13/110 *rows (key kept* 95/110*) — the churn maps are non-identical (by counterexample), with no known quotient.*

(2) *(halt predicate) the halt outcome is not a function on the abstract states: the first era-opening anchor of* $cm26$*'s own orbit,* $(5, 4, 2, 2)$ *at* pos 0 *(step 134), evolves normally for* $cm26$ *but halts for* $cm16$ *at step* 170*,* $cm51$ *at* 990*,* $cm52$ *at* 6294 *(reproduced by two independent engines). Halting therefore does not transfer through the era correspondence; non-halting proofs are assembled per machine.*

(3) *(large-era alignment is a trajectory phenomenon) over 562 aligned pairs with* $R_0 \geq 500$ *the offsets are vector-valued with flips (see the remark above) — a trajectory phenomenon, not a consequence of the drain-layer identity (which constrains only the within-era bookkeeping).*

Remark 5.4 (summary of the layered facts)**.** *The following are layered facts, not an isomorphism theorem (the four transition tables differ; no transition-level isomorphism exists):*

(1) *the net bookkeeping* $(+1, -2, +1)$ *is the same on all four machines (Lemma 3.2 machine-specific words + the* 166,976*-complete-era census identity), the intra-era drain maps are identical on the shared domain* $R \geq 4$ *(Theorem 5.2,* $cm26$*'s explicit odd-R correction), while the lap event words are machine-specific (Remark 3.3) and the churn/halting layers do not correspond (Proposition 5.3);*

(2) *the large-era alignment is a vector-offset trajectory phenomenon (Proposition 5.3(iii);* 3×10^9*-step census);*
(3) *end configurations agree bit-for-bit only on the frozen skeleton; working triples differ by* ± 100 *(a correction of the earlier "bitwise identical" measurement);*
(4) *the halt criteria have different sources (Theorems 2.1–2.4); non-halting assembly proceeds independently per machine (Proposition 5.3(ii)).*

6. The island invariant and the danger pattern

Definition 6.1 (island)**.** *An island is the pattern* $(0^1, 1^1, 0^1)$ *(a single-cell* 1*-island). A danger event is an occurrence of this pattern with the* ***head far*** *(not co-located with the destroyer wave).*

Theorem 6.2 (island invariant)**.** Verified domain: cm16 alone, 2×10^7 steps (near step 10^9), 892 island moments *[casecheck-island, 39 assertions; the checker's machine constant is cm16]. Within this domain: every island occurrence is co-located with the destroyer wave (head distance* ≤ 1*); island occurrences are transient — within each lap islands exist in* two *segments (correcting the earlier "only the boundary* $R{=}1$ *segment"), with creator and destroyer the same passing wave and existence window shorter than one lap. The cross-machine evidence is the audit deep scan (*10^9 *steps* $\times 4$*: head-far five-cell counts islFar* $= 0$*, dangers* $= 0$*; head-near counts 208,450/85,307/85,307/85,314) [quad6.exe audit; see dossier §5c].*

Definition 6.3 (vocabulary, audit and guard)**.** *The* verified vocabulary *is the set of routine branches actually occurring and catalogued in deep scans and censuses, consisting of three families: (a) the standard lap word with its parameter branches (Lemma 3.2,* $R \geq 2$*); (b) the drain branches (the wave-departure scripts for* $R \in \{1, 2, 3, 4\}$*); (c) the churn reseed branches. The branch index is the era key ((position class, neighborhood block count)), with measured key sets of 78/78/78/89 for the four machines (cm52 including 11 right-edge keys); the per-key parameterized output tables are authoritative in the census/era data and the classifier of* `quad.c` *(rerun entries at the end of the paper).* Audit mode*: at every A-state landing, check whether the danger pattern (a head-far* $(0^1, 1^1, 0^1)$*) appears in the five-cell window (head neighborhood* ± 2*), stepping to a given depth.* Guard mode*: at every step, check directly whether the machine's criterion (the two-cell condition of Theorems 2.1–2.4) holds, stepping to a given depth.*

Theorem 6.4 (safety within the vocabulary)**.** *At every A-state landing covered by the verified routine vocabulary, the pattern preconditions of Theorems 2.1–2.4 do not occur: sprint-landing right gaps* ≥ 2*; island positions excluded by Theorem 6.2; landing windows of the drain branches checked cell-by-cell against the census.*

Proof. Three branches are checked. (i) Sprint branch: the landing geometry is given by the word structure — the terminal event of a dance-family word places the head on the second cell of the new working gap with at least two 0s to its right, so the two-cell criterion fails; this is a verbatim property of the word structure (entry-enumeration of `casecheck-halt`, $\times 4$ machines). (ii) Island branch: Theorem 6.2 (cm16 transition-level + orbit audit; four-machine audit deep scan islFar$= 0$). (iii) Drain branch: the landing five-cell windows of the drain branches were checked instance-by-instance on the census key grid (78/78/78/89 keys $\times$ parameter grid), zero danger hits [casecheck/census data]. The three branches exhaust the vocabulary families. □

7. Reduction and open problem

Theorem 7.1 (reduction, per machine)**.** *For each machine* $m \in \{16, 26, 51, 52\}$ *the following are equivalent:*

(1) *m halts;*
(2) *the criterion pattern of m (the two-cell condition of Theorems 2.1–2.4) holds at an* out-of-vocabulary *landing (a danger event).*

Machine verification: audit mode steps through five-cell checks at out-of-vocabulary landings (10^9 *steps* $\times 4$*) with zero danger occurrences; guard mode (*2×10^{11} *steps* $\times 4$*, stepwise halt-precondition checking) with zero triggers; bare-machine deep scan of cm16 to* 5×10^{11} *steps with zero halts.*

Proof. (2)⇒(1): the criterion pattern triggers the in-edge chain of Theorems 2.1–2.4; the machine halts. (1)⇒(2): suppose a halt; then some criterion pattern is visited; by Theorem 6.4 (vocabulary safety) the visit occurs at an A-state landing outside the vocabulary coverage; the enumeration completeness of out-of-vocabulary landings is carried by the census/era key coverage (a finite closure observed in deep scans: the last new key of the four machines appears at steps 274/156/164/263 respectively, with zero new keys over the subsequent 10^9 steps — an empirical fact, not a theorem). □

Conjecture 7.2 (five-cell avoidance)**.** *No danger event ever occurs in the runs of the four machines (formally: clause (2) of Theorem 7.1 is false for all time).*

Remark 7.3 (radius-ladder experiment on the local abstraction (an observation, not a theorem))**.** *In the radius-ladder experiment for the key-set successor closure of cm16 (*$r = 2..6$*), $|K^*|$ grows by about* $\times 3.1$ *per added radius (*$99 \to 294 \to 939 \to 2863 \to 8879$*; two reruns reproduce digit-for-digit), with danger-key hits growing in step (*$2 \to 96$*) [kclosure].* Observation*: over the measured range the closure grows without bound in the radius. The experiment stops at* $r = 6$*; the statement "no finite fixed point exists for fixed-radius local abstraction" is a* conjecture *for* $r > 6$ *(no growth argument proved).*

Research field: piecewise-affine trajectory avoidance of the same shape as the cm2/M1 residuals (classification by analogy; per-machine formal restatement not performed).

Reproduction

`node casecheck-halt.mjs && node casecheck-island.mjs && node kclosure.mjs && node quad-equiv.mjs` (the last is the era-level correspondence experiment of Theorem 5.2/Proposition 5.3); `quad6.exe <code> audit 1000000000`; `quad6.exe <code> guard 200000000000`. All one-line commands.